SUPER SOFT LTD.

Project *Architecture Map*

Complete breakdown of every feature, page, endpoint, and data model.
Frontend · Admin Dashboard · Backend API · Database

Frontend
Public Website · Port 3000
15 pages

Admin Dashboard
Internal Panel · Port 3001
12 modules

Backend API
Express REST · Port 5000
70+ endpoints

Database
MongoDB · Port 27017
16 collections

Generated 13 June 2026

01 Frontend — Public Website

Next.js 14 · TypeScript · Tailwind CSS · Port 3000

PAGES & WHAT EACH CONTAINS

URL ROUTE	PAGE NAME	CONTENT & FEATURES
/	Home	Hero banner, animated stats, 4-service grid, tech stack showcase, 4-step process, client testimonials, CTA
/about	About	Company story, mission & vision, team cards (live from admin), awards (live from admin), core values
/services	Services Hub	Overview grid with links to all 5 service pages
/services/web-development	Web Dev	Custom web apps, e-commerce, SaaS — tech stack grid, process timeline, CTA
/services/mobile-app-development	Mobile Apps	Flutter & React Native, UI design, backend — tech stack, process, CTA
/services/ai-automation	AI & Automation	LLM integrations, workflow automation, chatbots, scraping — tech stack, process, CTA
/services/data-services	Data Services	Data entry, research, lead gen, mining, CRM — capabilities list, process, CTA
/services/data-scraping	Data Scraping	Web scraping, price monitoring, large-scale aggregation — tech stack, process, CTA
/services/chrome-extension	Chrome Extensions	Productivity tools, data capture, UI injection, OAuth — tech stack, process, CTA
/case-study	Portfolio	8 case studies, filter tabs (All / Web / Mobile / AI / Data / Scraping / Chrome), animated grid
/case-study/[slug]	Case Study Detail	Client bio, challenge, 4-step discovery, solution, key features, tech stack, 4-phase timeline, results, testimonial
/career	Careers	Job listings (live from admin), detail modal, application form with CV upload, company perks
/schedule-call	Schedule Call	Booking form !' saves to DB + auto-creates Meeting in admin + sends email confirmation
/privacy	Privacy Policy	10 sections: data collection, use, storage, sharing, cookies, user rights, retention, contact
/terms	Terms of Service	11 sections: acceptance, payment, IP, confidentiality, warranties, termination, governing law

01 Frontend — continued

UI Components · Library / Data Layer

REUSABLE UI COMPONENTS

COMPONENT FILE	WHAT IT RENDERS / DOES
layout/Navbar.tsx	Sticky navigation bar — logo, page links, "Schedule a Call" button
layout/Footer.tsx	Footer — service links, company info, social icons, copyright
home/HeroSection.tsx	Full-screen hero — headline, sub-text, two CTA buttons, floating shapes
home/StatsSection.tsx	Animated number counters: projects delivered, clients, years in business
home/ServicesSection.tsx	4-column cards linking to each service detail page
home/TechStackSection.tsx	Logo grid of all technologies the company works with
home/ProcessSection.tsx	4-step visual: Discovery !’ Design !’ Build !’ Launch
home/TestimonialsSection.tsx	Scrolling testimonials from past clients
home/CtaSection.tsx	"Ready to start?" CTA with schedule-call button
ui/ServiceCard.tsx	Reusable card: icon, title, description, arrow link
ui/Badge.tsx	Small pill badge for status/category labels
ui/AnimatedCounter.tsx	Counts up from 0 to target value when scrolled into view
ChatWidget.tsx	Floating chat bubble (bottom-right) — visitor enters name & email, sends messages, receives admin replies in real time via polling

DATA & API UTILITIES (FRONTEND/LIB/)

FILE	EXPORTS / RESPONSIBILITY
api.ts	scheduleCall(data) !’ POST /api/schedule | submitContact(data) !’ POST /api/contact | getJobs() !’ GET /api/careers | applyForJob(fd) !’ POST /api/careers/apply
case-studies.ts	CaseStudy type | caseStudiesData[] — 8 detailed case studies (FinTrack SaaS, ShopFlow App, AutoLead AI, MediCore Data, EduPortal LMS, SentimentAI, PriceWatch Scraper, TabBoost Extension) | badgeBg color map

02 Admin Dashboard

Next.js 14 · TypeScript · Tailwind CSS · Port 3001

Admin: admin@supersoftltd.com / SuperSoft@2024 Staff: staff@supersoftltd.com / Staff@2024

MODULES & CAPABILITIES

MODULE (ROUTE)	FEATURES & WHAT YOU CAN DO
/dashboard	8 KPI cards: revenue, projects, clients, tasks, messages, meetings, completed, invoices. Revenue bar chart (6 months). Projects-by-status pie chart. Recent projects table. New booking notification banner
/dashboard/clients	Full CRUD — name, company, email, phone, country, status (active / inactive / prospect), budget & paid totals. Search and filter bar
/dashboard/projects	Full CRUD — title, client, status (planning / in-progress / review / completed / on-hold), progress % slider, dates, budget & paid, tech stack tags, assigned team members
/dashboard/team	Staff profile cards with photo upload. Fields: name, role, email, phone, bio, skills, join date, salary. Salary history log (amount + date + note). Monthly payroll totals
/dashboard/invoices	Create with line items (desc, qty, unit price). Tax rate, discount. Payment: method (bank / card / Wise / PayPal / cash), paid amount, balance remaining. PDF download. Auto-email to client on creation. Status: draft / sent / paid / overdue
/dashboard/messages	Contact form submissions — split panel (list left, detail right). Unread badge. Mark as read, delete, reply via email
/dashboard/meetings	Meeting scheduler (title, date, duration, platform). Website bookings auto-appear with animated NEW badge + full visitor detail panel (name, email, phone, company, service, budget, message). Reply-by-email shortcut
/dashboard/jobs	Job postings CRUD — title, department, location, type (Full-time / Part-time / Contract). Toggle visibility. View incoming applications with CV downloads
/dashboard/case-studies	Full CRUD for all case study fields: challenge, discovery steps, solution, features, tech stack, timeline, results metrics, testimonial. Published toggle
/dashboard/chat	360px conversation panel — all visitor sessions, unread counts, open/closed badges. Right panel: full message thread, reply input, close button. Auto-polls every 4s
/dashboard/assistant	Claude AI chat — SSE streaming responses. File tools: readFile, writeFile, listDirectory (sandboxed). API key setup panel. Key stored in MongoDB — no .env needed
/dashboard/settings	Company name, email, phone, social links. Site stats editor: projects count, clients, team size, years — powers the public homepage counters
/dashboard/awards	Awards CRUD — title, organisation, year, description, category. Active toggle. Active awards shown on public About page automatically

02 Admin Dashboard — continued

Shared Components · Access Control

SHARED LAYOUT COMPONENTS

COMPONENT	PURPOSE
Sidebar.tsx	Left navigation rail — all module links, live badge counters (chat unread, new meeting bookings). Polls backend every 10 s. Hides admin-only items from staff role
Header.tsx	Top page bar — page title + optional action button (e.g. "+ New Invoice")
StatCard.tsx	KPI tile on overview dashboard — label, large value, icon, accent colour
Modal.tsx	Reusable slide-in modal used for all CRUD forms across every module

ROLE-BASED ACCESS CONTROL

ROLE	WHAT THEY CAN ACCESS
Admin	Full access — all 13 modules including team salaries, system settings, Claude AI assistant, and case study management
Staff	Restricted — can view clients, projects, tasks, and live chat; cannot manage invoices, salaries, settings, jobs, or AI assistant

03 Backend API

Express.js + TypeScript · Port 5000 · MongoDB via Mongoose

PUBLIC ENDPOINTS (NO AUTHENTICATION NEEDED)

METHOD	ENDPOINT	WHAT IT DOES
GET	/api/health	Health check — returns { status: 'ok', uptime }
POST	/api/schedule	Book a call: saves ScheduleCall !’ creates Meeting (source=website, isNew=true) !’ emails admin. Rate-limited
POST	/api/contact	Save contact form submission !’ email admin. Rate-limited
GET	/api/careers	List active job postings (isActive=true) for public careers page
POST	/api/careers/apply	Submit application — multipart, CV saved via Multer to /uploads/cv/
GET	/api/case-studies/public	Published case studies for public portfolio
GET	/api/case-studies/public/:slug	Single case study by slug (for detail page)
GET	/api/team/public	Active team members for public About page
GET	/api/awards/public	Active awards for public About page
GET	/api/site-stats	Homepage statistics (projects, clients, years)
POST	/api/chat/start	Start visitor chat !’ returns unique sessionId (stored in browser localStorage)
POST	/api/chat/:sessionId/message	Visitor sends a message
GET	/api/chat/:sessionId/messages	Visitor polls for admin replies (every 4s)
GET	/api/assistant/status	Returns { configured: boolean } — is Claude API key set?

03 Backend API — continued

Protected Endpoints (JWT Bearer token required)

AUTH

METHOD	ENDPOINT	WHAT IT DOES
POST	/api/auth/login	Verify email + bcrypt password ⁉ return signed JWT (7-day expiry) + role
GET	/api/auth/me	Return current user profile from token

CLIENTS · PROJECTS · TASKS

METHOD	ENDPOINT	WHAT IT DOES
GET	/api/clients	List all clients
POST	/api/clients	Create client
PUT	/api/clients/:id	Update client
DELETE	/api/clients/:id	Delete client
GET	/api/projects	List projects (client + team populated)
POST	/api/projects	Create project
PUT	/api/projects/:id	Update project
DELETE	/api/projects/:id	Delete project
GET	/api/tasks	List tasks (filter by projectId optional)
POST	/api/tasks	Create task
PUT	/api/tasks/:id	Update task status, priority, assignee
DELETE	/api/tasks/:id	Delete task

INVOICES · MEETINGS · MESSAGES

METHOD	ENDPOINT	WHAT IT DOES
GET	/api/invoices	List all invoices (client populated)
POST	/api/invoices	Create invoice — sendEmail:true auto-sends branded HTML + PDF to client
PUT	/api/invoices/:id	Update invoice
DELETE	/api/invoices/:id	Delete invoice
GET	/api/invoices/:id/pdf	Stream generated PDF to browser
POST	/api/invoices/:id/send-email	Re-send invoice email with PDF attachment
GET	/api/meetings	List all meetings (createdAt desc)
POST	/api/meetings	Create meeting manually
PUT	/api/meetings/:id	Update meeting

DELETE	/api/meetings/:id	Delete meeting
GET	/api/meetings/new-count	Count unread website bookings { count: N }
PUT	/api/meetings/:id/mark-read	Clear isNew flag on website booking
GET	/api/messages	All contact form submissions
PUT	/api/messages/:id/read	Mark as read
DELETE	/api/messages/:id	Delete message

03 Backend API — continued

Team · Jobs · Chat · Claude AI · Content

TEAM · JOBS · AWARDS · CASE STUDIES · SITE STATS

METHOD	ENDPOINT	WHAT IT DOES
GET	/api/team	All team members (admin — includes salary)
POST	/api/team	Create member
PUT	/api/team/:id	Update member
DELETE	/api/team/:id	Delete member
POST	/api/team/:id/salary	Log salary payment
POST	/api/team/upload-photo	Upload photo via Multer !' /uploads/team/
GET	/api/careers/jobs	All job postings (admin — includes hidden)
POST	/api/careers/jobs	Create job
PUT	/api/careers/jobs/:id	Update job
DELETE	/api/careers/jobs/:id	Delete job
GET	/api/awards	All awards (admin)
POST	/api/awards	Create award
PUT	/api/awards/:id	Update award
DELETE	/api/awards/:id	Delete award
GET	/api/case-studies	All case studies (admin — includes drafts)
POST	/api/case-studies	Create case study
PUT	/api/case-studies/:id	Update case study
DELETE	/api/case-studies/:id	Delete case study
PUT	/api/site-stats	Update homepage stats

LIVE CHAT · CLAUDE AI ASSISTANT

METHOD	ENDPOINT	WHAT IT DOES
GET	/api/chat/admin/conversations	All chat sessions with visitor info + unread counts
GET	/api/chat/admin/:sessionId/messages	Full message thread (admin view)
POST	/api/chat/admin/:sessionId/reply	Admin sends reply
PUT	/api/chat/admin/:sessionId/close	Close conversation
POST	/api/assistant/save-key	Save Claude API key to MongoDB AppSettings
DELETE	/api/assistant/save-key	Remove Claude API key from DB
POST	/api/assistant/chat	SSE streaming — proxies to Anthropic API, handles tool_use (readFile, writeFile, listDirectory)

04 Database — MongoDB

16 collections · DB name: supersoft · Port 27017

BUSINESS CORE

COLLECTION	FIELDS STORED
User	email, password (bcrypt), role (admin | staff), active, timestamps
Client	name, company, email, phone, country, status (active | inactive | prospect), notes, totalBudget, totalPaid, timestamps
Project	name, client !'Client, status (planning | in-progress | review | completed | on-hold), progress 0–100, startDate, deadline, budget, paid, stack[], description, team[] !'TeamMember, notes, timestamps
Task	title, project !'Project, assignee !'User, priority (low | medium | high | urgent), status (todo | in-progress | done), dueDate, description, timestamps
Invoice	invoiceNo (unique), client !'Client, project !'Project, items[] {desc, qty, unitPrice, amount}, subtotal, taxRate, tax, discount, total, paid, paymentMethod (bank | card | wise | paypal | cash | other), paymentNotes, status (draft | sent | paid | overdue), issueDate, dueDate, notes, emailSent bool, timestamps
ScheduleCall	name, email, company, phone, service, budget, message, timestamps !• raw submissions from /schedule-call page
Meeting	title, client !'Client, project !'Project, date, duration, platform, notes, status (scheduled | completed | cancelled), attendees[], source (website | manual), isNew bool, visitorName, visitorEmail, visitorPhone, visitorCompany, visitorService, visitorBudget, visitorMessage, timestamps
Contact	name, email, subject, message, read bool, timestamps !• contact form submissions

TEAM & CAREERS

COLLECTION	FIELDS STORED
TeamMember	name, role, email, phone, bio, skills[], salary, salaryHistory[] {amount, date, note}, joinDate, active bool, linkedin, github, photo (path in /uploads/team/), timestamps
Job	title, department, location, type (Full-time | Part-time | Contract | Freelance), description, requirements[], isActive bool, timestamps
JobApplication	name, email, phone, position, coverLetter, cvPath (path in /uploads/cv/), timestamps

COMMUNICATIONS

COLLECTION	FIELDS STORED
ChatConversation	sessionId (unique UUID), visitorName, visitorEmail, status (open | closed), unreadCount, lastMessage, lastMessageAt, timestamps
ChatMessage	conversationId (indexed) !'ChatConversation, sender (visitor | admin), senderName, message text, read bool, timestamps

CONTENT & CONFIGURATION

COLLECTION	FIELDS STORED
AdminCaseStudy	slug (unique), category, project, client, description, tags[], metrics[], gradient/border/accent, clientBio, challenge, discovery[] {step, detail}, solution, keyFeatures[], techStack[], timeline[] {phase, duration, tasks}, results[] {label, value, sub}, testimonial {quote, author, role}, published bool, timestamps

Award	title, organization, year, description, category, active bool, timestamps
SiteStats	projects, projectsSuffix, projectsSub, clients, clientsSuffix, clientsSub, team, teamSuffix, teamSub, years, yearsSuffix, yearsSub !• powers homepage animated counters
AppSettings	key (unique string), value (string) !• stores Claude API key entered via admin UI

How the Layers Connect

End-to-end data flows for the 5 most important interactions

1. Visitor books a call on /schedule-call

Frontend	Form submitted !’ POST /api/schedule
Backend	Saves ScheduleCall !’ creates Meeting (source=website, isNew=true) !’ emails admin
Database	ScheduleCall + Meeting documents written
Admin	Sidebar badge increments !’ Meetings page shows NEW pulse badge + visitor detail panel

2. Admin creates an invoice and emails client

Admin	Fill line items, toggle "Send to Client" !’ POST /api/invoices { sendEmail: true }
Backend	Calculates totals !’ saves Invoice !’ buildPdfBuffer() !’ sendInvoiceEmail() via Nodemailer SMTP
Database	Invoice saved with emailSent=true, status=sent
Client	Receives branded HTML email + PDF attachment from company Gmail

3. Visitor chats via the floating chat widget

Frontend	Chat bubble opened !’ POST /api/chat/start !’ sessionId saved to localStorage
Frontend	Message sent !’ POST /api/chat/:id/message !’ polls GET /api/chat/:id/messages every 4s
Admin	Conversation appears with unread badge !’ admin replies !’ POST /api/chat/admin/:id/reply
Database	ChatConversation + ChatMessage documents; unreadCount tracked

4. Admin uses Claude AI assistant

Admin	Types prompt !’ POST /api/assistant/chat
Backend	Reads key from MongoDB AppSettings !’ SSE stream from Anthropic !’ handles tool_use (readFile, writeFile, listDirectory)
Frontend	Tokens stream in real time — response appears as it's generated

5. Public About page shows live team & awards

Admin	Adds team member (with photo) or award !’ sets active=true
Database	TeamMember / Award documents updated
Backend	GET /api/team/public + /api/awards/public return only active records
Frontend	/about page fetches on load — team and awards sections update automatically